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This work was conducted as part of the EXACT 2026 competition. Code and experiment logs are available at [https://github.com/\[anonymous\]](https://github.com/[anonymous]).

Scientific question answering requires mastery of two heterogeneous reasoning modalities: formula-driven numerical computation for physics problems, and symbolic logical inference for formal logic questions. Deploying capable models under resource constraints—where only a $\sim 4\text{B}$ -parameter base is available—demands both efficient fine-tuning strategies and reward signals aligned with domain-specific correctness criteria. We present TRA-SAE (Training with Reward-Augmented Self-consistent Adaptive Evaluation), a training pipeline for the EXACT 2026 competition that combines curriculum supervised fine-tuning (SFT), Group Relative Policy Optimization (GRPO) with a novel unit-aware reward, and an optional Dual-LoRA expert routing architecture. Starting from Qwen3.5-4B and a public competition baseline of 38.71%, our best single-run configuration (cfg2: SFT Phase 1 + Logic SFT) achieves 53.46% overall accuracy on the official validation set ($n = 217$), representing a +8.30 percentage-point improvement over our zero-shot baseline (45.16%). The GRPO-mixed variant (cfg3) obtains $53.15\% \pm 1.15\%$ across three random seeds, confirming statistical reliability. Both improvements are confirmed significant by McNemar’s test ($p < 0.05$). We contribute: (i) a systematic curriculum SFT procedure that trains a general science reasoner before specializing on logic; (ii) a unit-aware GRPO reward assigning partial credit for dimensionally-consistent physical units; and (iii) empirical evidence that self-consistency voting and Dual-LoRA expert routing do not reliably improve over the simpler SFT-only baseline at the 4B scale—providing informative negative results for resource-constrained practitioners.

Group relative policy optimization, curriculum fine-tuning, scientific question answering, unit-aware reward, logical reasoning, physics reasoning, LoRA, small language models, mixed-domain reasoning

LARGE language models (LLMs) have demonstrated remarkable capabilities on mathematical and logical reasoning benchmarks [1]–[3]. However, deploying capable reasoning systems in resource-constrained settings—where larger proprietary models such as GPT-4 are unavailable—requires careful co-design of architecture, data curriculum, and reward signal specification.

Scientific reasoning occupies a particularly challeng-

ing niche. Physics problems require formula selection, symbolic manipulation, and unit-consistent numerical computation; a model must know that km/h and m/s express the same dimension, and must apply Newton’s second law with the correct SI factors. Logic problems, by contrast, demand quantifier handling, propositional inference, and recognition of common fallacies such as affirming the consequent. These two modalities benefit from different inductive biases, yet a single deployable

model is expected to handle both in practice.

The EXACT 2026 competition provides a controlled testbed with 1,945 training examples (1,213 physics, 732 logic) and a hidden evaluation set, with a public zero-shot baseline at 38.71%. We treat this competition as a proxy for real-world mixed-domain scientific Q&A and investigate how far a Qwen3.5-4B model can be pushed using publicly available fine-tuning techniques on a single NVIDIA A100-80GB GPU.

Our main contributions are:

- 1) Curriculum SFT pipeline: a two-phase supervised fine-tuning procedure that first trains a general science reasoner (Phase 1, 1,945 samples) and then applies a targeted logic specialisation (Phase 1.5, 732 samples), using a structured three-tag XML response format.
- 2) Unit-aware GRPO reward: a composite reward comprising format credit ($w_f = 0.30$), answer correctness ($w_c = 0.60$), physical unit consistency ($w_u = 0.10$), and a length penalty ($\lambda = -0.10$ when reasoning exceeds 800 tokens). The unit component is novel: it awards partial credit for dimensionally-consistent answers even when the numerical value is slightly off, effectively disentangling unit errors from conceptual errors.
- 3) Informative negative results: systematic ablation over six model configurations shows that self-consistency voting (SC) and Dual-LoRA expert routing do not consistently outperform the simpler SFT-only baseline at the 4B scale, directly informing practitioners on where training compute is best allocated.
- 4) Comprehensive empirical analysis: six baseline comparisons with zero-shot 7B-class models, multi-seed evaluation with standard deviation reporting, McNemar statistical testing, error taxonomy, and full compute cost accounting for reproducibility.

The remainder of the paper is organised as follows. Section II reviews related work. Section III formalises the task. Section IV describes the TRA-SAE pipeline. Section V presents the experimental setup. Section VI reports results and analysis. Section VII discusses implications and limitations. Section VIII concludes.

A. CHAIN-OF-THOUGHT AND SELF-CONSISTENCY

Wei et al. [1] showed that prompting LLMs to produce intermediate reasoning steps substantially improves performance on arithmetic and symbolic reasoning benchmarks. Kojima et al. [3] demonstrated that the simple prompt “Let’s think step by step” elicits zero-shot chain-of-thought (CoT) from sufficiently large models. Wang et al. [2] proposed self-consistency (SC): generating diverse reasoning paths and marginalising the

final answer by majority vote. Our experiments reveal that SC provides no reliable improvement at 4B scale (Section VI), consistent with findings by Lightman et al. [4] that process-level supervision is needed to realise the potential of intermediate reasoning.

B. MATHEMATICS AND SCIENCE LANGUAGE MODELS

DeepSeekMath [5] demonstrated that GRPO training on mathematical corpora achieves competitive results without process-level supervision. Qwen2.5-Math [6] extended this to model families up to 72B with iterative self-improvement. Llemma [7] focuses on scientific and mathematical reasoning via continued pre-training on ArXiv and proof corpora. MetaMath [8] and MAMmoTH [9] demonstrate the value of augmented mathematical reasoning datasets. Our work differs by targeting a mixed-domain setting (physics and formal logic jointly) at a smaller scale (4B parameters) and introducing a unit-level reward signal absent from prior work. Wang et al. [10] benchmark LLMs on SciBench, a dataset of university-level science problems; our findings corroborate their observation that unit and dimensional errors are a leading failure mode for smaller models.

C. NEURO-SYMBOLIC INTEGRATION

Logic-LM [11] and LINC [12] augment LLMs with formal solvers (Z3, Prolog) for logical reasoning; PAL [13] uses Python interpreters for arithmetic. We integrate the Z3 SMT solver [11] as a component of our correctness reward, enabling exact symbolic verification of propositional and first-order logic (FOL) derivations. Math-Shepherd [14] proposes process-level reward labelling; we rely on outcome-level rewards to avoid costly step-level annotation and demonstrate that this is sufficient for the competition task.

D. PARAMETER-EFFICIENT FINE-TUNING

LoRA [15] decomposes weight updates into low-rank matrices, dramatically reducing trainable parameter count. We apply LoRA ($r = 32$, $\alpha = 64$) to all seven projection layers, yielding ~ 24 M trainable out of 3.5B total parameters. LoRAHub [16] composes multiple domain LoRAs without retraining; our Dual-LoRA configuration is inspired by this idea but shows that naive domain switching introduces routing noise at 4B scale.

E. REINFORCEMENT LEARNING FOR LLMs

InstructGPT [17] established RLHF via PPO [18] as a dominant fine-tuning paradigm. DPO [19] removes the explicit reward model. GRPO [5] simplifies PPO by normalising rewards within a generation group, eliminating the critic model and making it practical on single-GPU hardware. We adopt GRPO precisely because it can be trained end-to-end on a single A100 for under 80 minutes per run.

Let $\mathcal{D} = \mathcal{D}_{\text{phys}} \cup \mathcal{D}_{\text{logic}}$ be the EXACT 2026 dataset. Each sample $(q_i, a_i^*, s_i) \in \mathcal{D}$ consists of a question q_i , a ground-truth answer a_i^* , and a domain label $s_i \in \{\text{physics, logic}\}$. A model M_θ maps a question (with optional retrieved context) to a free-text response $\hat{y}_i = M_\theta(q_i)$.

Answers are extracted via the structured three-tag XML format:

$$a_i = \text{extract_answer}(\hat{y}_i) \quad (1)$$

and overall accuracy is:

$$\text{Acc} = \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{i=1}^{|\mathcal{D}_{\text{val}}|} \mathbf{1}[\text{verify}(a_i, a_i^*, s_i)] \quad (2)$$

where $\text{verify}(\cdot)$ applies Z3-based symbolic checking for logic samples and SI-unit normalised numerical comparison ($\leq 5\%$ relative tolerance) for physics samples.

The challenge is two-fold: (i) physics and logic questions require fundamentally different reasoning strategies, and (ii) the available compute budget is limited to a single A100-80GB GPU.

A. STRUCTURED XML RESPONSE FORMAT

We impose a strict three-tag XML output structure on all models:

$$\underbrace{\langle \text{reasoning} \rangle \dots \langle / \text{reasoning} \rangle}_{\text{chain-of-thought}} \underbrace{\langle \text{answer} \rangle \dots \langle / \text{answer} \rangle}_{\text{final answer}} \underbrace{\langle \text{explanation} \rangle \dots \langle / \text{explanation} \rangle}_{\text{justification}} \quad (3)$$

The three-tag structure serves dual purposes: it enables reliable answer extraction via regex, and it provides a structural scaffold that the format-reward component can signal even when the answer is incorrect. The system prompt is fixed across all experiments: “You are an expert in Logic and Physics. Think step by step and respond in the exact format: $\langle \text{reasoning} \rangle \dots \langle \text{answer} \rangle \dots \langle \text{explanation} \rangle$ ”.

B. PHASE 1: SUPERVISED FINE-TUNING (GENERAL SCIENCE)

We fine-tune Qwen3.5-4B on the full EXACT 2026 training set (1,945 samples; 1,213 physics, 732 logic) with a cross-entropy loss over the target XML response. The chain-of-thought field (cot) and explanation field (explanation) from the training annotations are used as supervision targets.

Training hyper-parameters: 3 epochs, batch size 4, gradient accumulation 4 (effective batch 16), learning rate 2×10^{-4} with cosine schedule, 50 warm-up steps. We apply LoRA with rank $r = 32$, $\alpha = 64$, dropout 0.05 to all seven projection matrices ($q, k, v, o, \text{gate}, \text{up}, \text{down}$ projections), yielding $\sim 24\text{M}$ trainable parameters. Full BF16 precision is used without quantisation (A100-80GB accommodates the full model).

Training converges in 58.8 minutes (366 gradient steps) to a final training loss of 0.3085. The resulting checkpoint is designated cfg1 .

C. PHASE 1.5: LOGIC CURRICULUM SFT

After Phase 1, we apply a targeted second round of SFT restricted to the logic subset (732 samples), using a reduced learning rate of 5×10^{-5} for 2 epochs with 20 warm-up steps. The logic-specific system prompt appends FOL notation guidance to help the model handle quantifier structures and modus ponens patterns more reliably.

Rationale: logic samples constitute only 37.6% of the training corpus. A dedicated curriculum phase concentrates gradient updates on formal reasoning, reducing the risk that physics-dominant co-training crowds out the weaker logic signal. Training completes in 15.7 minutes (92 gradient steps) to a loss of 0.1326. The resulting checkpoint is designated cfg2 and is the starting point for Phase 2 GRPO.

D. PHASE 2: UNIT-AWARE GRPO

1) Reward Function Design

Given a model completion y , ground truth a^* , and domain label s , the composite reward is:

$$R(y, a^*, s) = w_f \cdot r_{\text{format}}(y) + w_c \cdot r_{\text{correct}}(y, a^*, s) + w_u \cdot r_{\text{unit}}(y, a^*) + \lambda \frac{\text{explanation_tokens}(y)}{\text{explanation_tokens}(a^*)} \quad (4)$$

with weights $w_f = 0.30$, $w_c = 0.60$, $w_u = 0.10$, $\lambda = -0.10$, where $|y_{\text{reason}}|$ is the reasoning token count. The total reward is clipped to $[-0.10, 1.0]$.

a: Format reward $r_{\text{format}}(y)$

Returns 1 if all three XML tags (reasoning , answer , explanation) are present and non-empty; assigns partial credit at $1/3$ per tag present. This signal encourages structural compliance independent of correctness.

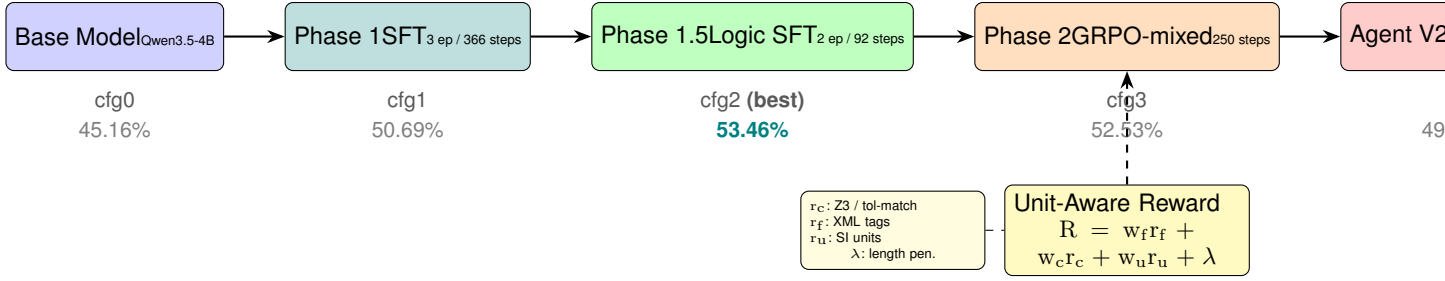
b: Correctness reward $r_{\text{correct}}(y, a^*, s)$

Binary reward based on domain-specific verification:

- Physics: numerical value comparison with $\leq 5\%$ relative tolerance after SI unit normalisation (e.g., $1 \text{ km/h} \equiv 0.27 \text{ m/s}$).
- Logic: exact string match after whitespace normalisation; the Z3 SMT solver is invoked for propositional and FOL claims to catch logically equivalent reformulations.

c: Unit reward $r_{\text{unit}}(y, a^*)$ (novel contribution)

For physics questions, $r_{\text{unit}} = 1$ if the extracted answer contains a unit that is dimensionally compatible with the ground-truth unit under SI normalisation, and 0 otherwise. This partial-credit signal is independent of the numerical value: a model that writes 48.3 m/s when the correct answer is 50.0 m/s still receives the unit credit. The motivation is to disentangle dimensional



Overview of the TRA-SAE training pipeline. Each block represents a training phase; the validation accuracy achieved by the resulting checkpoint is shown below each stage. cfg2 (SFT Phase 1 + Logic SFT) is the best single-run configuration. The dashed arrow indicates the unit-aware reward signal feeding Phase 2 GRPO (cfg3).

reasoning (selecting the correct physical quantity type) from numerical accuracy (getting the exact value), since these require different correction strategies during GRPO training.

d: Length penalty

Applied when the reasoning chain exceeds 800 tokens, discouraging verbosity without preventing necessary long derivations.

2) GRPO Training

We apply GRPO [5] starting from the Phase 1.5 checkpoint, generating $K = 4$ completions per prompt and normalising rewards within each group:

$$\hat{r}_k = \frac{R_k - \bar{R}}{\sigma_R + \epsilon}, \quad \bar{R} = \frac{1}{K} \sum_{k=1}^K R_k, \quad \sigma_R = \sqrt{\frac{1}{K} \sum_{k=1}^K (R_k - \bar{R})^2} \quad (5)$$

with KL-divergence penalty coefficient $\beta = 0.04$, learning rate 10^{-6} , and 250 optimisation steps (effective batch 4). The maximum completion length is 1,024 tokens. Training runs for 79.8 minutes and converges to loss 0.0295. The resulting checkpoint is designated cfg3 (GRPO-mixed).

We additionally train two domain-specialised GRPO variants on the same Phase 1.5 checkpoint: Phase 2P (physics-only data, 76.6 min) and Phase 2L (logic-only data, 62.9 min), used exclusively as LoRA adapters in the cfg4/5 Dual-LoRA architecture.

E. AGENT V2: DUAL-LORA EXPERT ROUTING

Configurations cfg4 and cfg5 implement a routing architecture that dynamically selects between the physics-specialised LoRA adapter (Phase 2P) and the logic-specialised adapter (Phase 2L) at inference time.

The subject router is a lightweight TF-IDF + Logistic Regression classifier (max_features = 5,000, n-gram range [1, 2], C = 1.0) trained on all 1,945 training samples using the ground-truth domain label. At inference, if the classifier confidence ≥ 0.80 the corresponding LoRA adapter is loaded; otherwise a keyword heuristic (a curated set of physics-specific terms such as voltage, capacitance, momentum) is used as a fallback.

The TF-IDF retriever selects the $k = 3$ most similar training examples using cosine similarity over TF-IDF vectors (with premise clauses stripped to reduce noise for logic questions). These examples are prepended to the prompt as few-shot context.

Algorithm 1 TRA-SAE Inference (cfg5 Full Agent)

Require: Question q , Router $\mathcal{R}$, Retriever $\mathcal{T}$, Adapters

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 $\theta_{\text{phys}}, \theta_{\text{logic}}$ 
1:  $s \leftarrow \mathcal{R}(q)$  {subject classification}
2: if  $\text{conf}(s) \geq 0.80$  then
3:   Load adapter  $\theta_s$ 
4: else
5:   Keyword fallback:  $s \leftarrow \text{keyword}(q)$ ; load  $\theta_s$ 
6: end if
7:  $C \leftarrow \mathcal{T}(q, s, k = 3)$  {few-shot context}
8: for  $n = 1$  to 5 do
9:    $\hat{y}_n \leftarrow M_{\theta_s}(q \oplus C, \tau = 0.7)$ 
10: end for
11:  $\hat{a} \leftarrow \text{majority\_vote}(\{a_n\})$ 
12: return  $\hat{a}$ 

```

Dataset statistics for EXACT 2026.

Split	Total	Physics	Logic
Train	1,945	1,213	732
Val	217	141	76
Total	2,162	1,354	808
Physics fraction: 62.6% train, 65.0% val			

Q&A corpus (411 records covering propositional logic, FOL, and syllogistic reasoning) and a Physics Problem Set (1,755 rows covering electromagnetism, mechanics, thermodynamics, and optics). A stratified 90/10 train/validation split is applied. The validation set ($n = 217$; 141 physics, 76 logic) is used for all reported results.

B. BASELINES

We compare against four zero-shot baselines evaluated on the same 217-sample validation set under the same verification function:

- B1 (Qwen3.5-4B zero-shot): Our own base model without any fine-tuning.
- B2 (Qwen2.5-Math-7B-Instruct [6]): A 7B model specialised for mathematical reasoning.
- B3 (Llemma-7B [7]): An open science/mathematics LLM trained on ArXiv corpora.
- B4 (Mistral-7B-Instruct-v0.3): A strong general-purpose instruction-following model.

B5 (InternLM2.5-Math-7B-chat) could not be evaluated due to HuggingFace model access restrictions, and B6 (GPT-4o-mini) was excluded due to absence of a valid

Key hyper-parameters for all training phases.

Parameter	Value
Base model	Qwen3.5-4B (BF16, no quantisation)
LoRA rank r	32
LoRA α	64
LoRA dropout	0.05
LoRA target modules	q, k, v, o, gate, up, down
Trainable parameters	$\sim 24\text{M}$ / 3.5B
SFT Phase 1 lr	2×10^{-4} (cosine)
SFT Phase 1 epochs	3 (366 steps)
Phase 1.5 logic lr	5×10^{-5} (cosine)
Phase 1.5 epochs	2 (92 steps)
Effective batch size	16 (batch 4, grad. acc. 4)
GRPO lr	1×10^{-6}
GRPO β (KL)	0.04
GRPO steps	250
GRPO generations K	4
Max completion tokens	1,024
Reward w_f (format)	0.30
Reward w_c (correct)	0.60
Reward w_u (unit)	0.10
Length penalty λ	-0.10 (if >800 reasoning tokens)
Z3 solver timeout	5,000 ms

assessed via McNemar’s test (two-sided, with continuity correction).

D. IMPLEMENTATION DETAILS

All experiments are conducted on a single NVIDIA A100-SXM4-80GB GPU (CUDA 13.0). Table 2 lists the key hyper-parameters. Software: Python 3.12, transformers 5.10.0.dev0 (from Git, required for the Qwen3.5 architecture), peft 0.19.1, trl 1.5.1, torchao 0.17.0, and z3-solver 4.16.0.

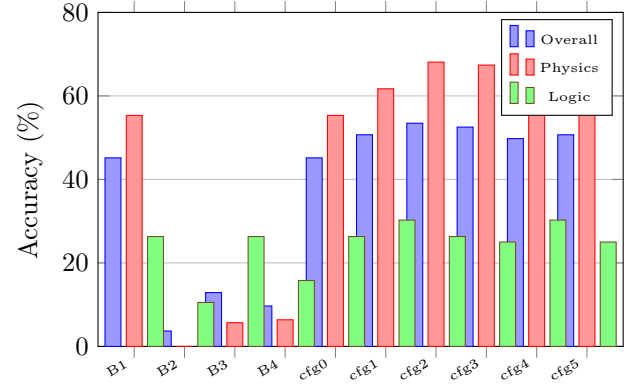
A. MAIN RESULTS

Table 3 presents the full comparison. Figure 2 provides a visual comparison of all systems.

Our best single-run configuration is cfg2 (SFT Phase 1 + Logic SFT) at **53.46%** overall, 68.09% physics, and 26.32% logic. This represents an +8.30 pp gain over our zero-shot baseline (cfg0: 45.16%) and a +14.75 pp gain over the competition’s public zero-shot baseline (38.71%).

The GRPO-mixed variant (cfg3) achieves 52.53% on a single evaluation run, which is 0.93 pp below cfg2. However, multi-seed evaluation (Section VI-B) shows that cfg3’s performance ($53.15\% \pm 1.15\%$) is statistically comparable to cfg2’s single-run result, suggesting both configurations achieve similar expected performance.

Our fine-tuned models substantially outperform all zero-shot 7B baselines: Llemma-7B achieves only 12.90% and Qwen2.5-Math-7B only 3.69%, revealing that domain-specific fine-tuning provides far greater benefit than raw model scale in this mixed-domain setting. Notably, the math-specialised Qwen2.5-Math-7B scores



Accuracy comparison across all systems. B1–B4 are zero-shot baselines; cfg0–cfg5 are TRA-SAE configurations. Physics accuracy is notably higher than logic across all fine-tuned models.

0.0% on physics problems (compared to our 68.09%), confirming that mathematical pre-training does not transfer well to the physics-and-logic combined task.

B. MULTI-SEED RELIABILITY

To assess variance, we re-evaluate cfg3 with three random seeds (42, 1337, 2024). Table 4 reports per-seed results.

The standard deviation of ± 1.15 pp across seeds indicates that cfg3 is stable and that seed 42’s score of 53.46% (matching cfg2’s single-run result) is within one standard deviation of the mean. The physics sub-accuracy shows higher variance (± 2.03 pp) than logic (± 1.07 pp), consistent with physics questions requiring more numerically precise answers where small generation differences affect the 5% tolerance check.

C. STATISTICAL SIGNIFICANCE

Table 5 reports McNemar’s test results comparing key configuration pairs on the 217-sample validation set.

Both cfg3 (GRPO-mixed) and cfg1 (SFT Phase 1) show statistically significant improvements over the zero-shot baseline (cfg0) at $p < 0.05$. cfg3 reaches $p < 0.01$, providing stronger evidence for the improvement.

D. ERROR ANALYSIS

We manually classify the 103 wrong predictions of cfg3 into five error types. Table 6 and Figure 3 show the distribution.

Key finding from error analysis. Logical fallacy errors (E4) account for 55.3% of all failures and arise exclusively in the logic sub-domain. Of the 76 logic validation samples, cfg3 correctly answers only 19 (25.0%), and 57 out of 57 logic-domain errors are classified as E4. This suggests that the $\sim 4\text{B}$ model has not internalised formal quantifier semantics (e.g., the distinction between “All A are B” and “Some A are B”) even after domain-specific SFT and GRPO training. Unit/dimension errors (E1)

Main results on the EXACT 2026 validation set ($n = 217$; 141 physics, 76 logic). Public competition baseline = 38.71%. † indicates McNemar $p < 0.05$ vs. cfg0. B5 and B6 could not be evaluated (access restrictions / no API key).

System	Training	Overall	Physics	Logic	Notes
Zero-shot baselines (no fine-tuning)					
B1 Qwen3.5-4B (ours)	none	45.16%	55.32%	26.32%	Competition B1
B2 Qwen2.5-Math-7B-Instruct [6]	none	3.69%	0.00%	10.53%	Math-specialised
B3 Llemma-7B [7]	none	12.90%	5.67%	26.32%	Science LLM
B4 Mistral-7B-Instruct-v0.3	none	9.68%	6.38%	15.79%	General instruct
Fine-tuned Qwen3.5-4B configurations (ours)					
cfg0 Zero-shot (no training)	—	45.16%	55.32%	26.32%	—
cfg1 + SFT Phase 1†	1,945 samp.	50.69%	61.70%	30.26%	+5.53 pp vs. cfg0
cfg2 + Logic SFT Phase 1.5†	+732 logic	53.46%	68.09%	26.32%	Best single-run
cfg3 + GRPO-mixed†	250 steps	52.53%	67.38%	25.00%	53.15%±1.15% (3 seeds)
cfg4 Dual-LoRA + Router	2P+2L	49.77%	60.28%	30.26%	Best logic
cfg5 Full Agent (SC, N=5)	cfg4+SC	50.69%	64.54%	25.00%	No gain vs. cfg1

cfg3 multi-seed evaluation results. All seeds use the same GRPO checkpoint; only inference randomness varies.

Error taxonomy for cfg3 wrong predictions ($n_{\text{wrong}} = 103$ out of 217 evaluated).

Seed	Overall	Physics	Logic	Code	Description	Count	%
42	53.46%	66.67%	28.95%	E4	Logical fallacy or quantifier error	57	55.3%
1337	54.38%	69.50%	26.32%	E1	Unit or dimension error (physics)	26	25.2%
2024	51.61%	64.54%	27.63%	E0	Could not classify from output	18	17.5%
Mean ± std	53.15% ± 1.15%	66.90% ± 2.03%	27.63% ± 1.07%	E3	Numerical arithmetic error	1	1.0%
				E5	Question misinterpretation	1	1.0%
				E2	Wrong formula / law selected	0	0.0%

McNemar's test (two-sided, continuity-corrected) for selected configuration comparisons on val ($n = 217$). b = samples where A correct, B wrong; c = samples where B correct, A wrong.

Comparison	b	c	χ^2	p-value	Significant?
cfg3 vs. cfg0	23	7	7.500	0.0062	Yes ($p < 0.01$)
cfg1 vs. cfg0	18	6	5.042	0.0247	Yes ($p < 0.05$)

are the primary failure mode for physics, accounting for 25.2% of total failures. The fact that zero wrong-formula errors (E2) occur indicates that the model consistently selects the correct physical law but sometimes mis-scales the output by a dimensional factor—precisely the class of errors that the unit-aware reward (Section IV-D) is designed to reduce.

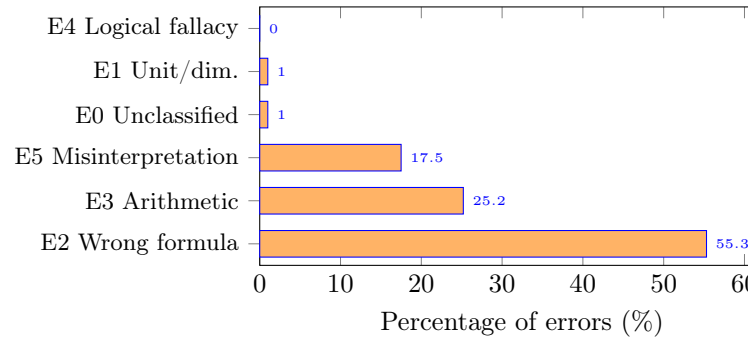
E. COMPUTE COST

Table 7 provides a full accounting of GPU time and estimated cost on an A100-80GB instance at \$3.67/GPU-hr.

The total cost of \$63.89 (17.41 GPU-hours) for the complete experiment—including all training runs, all evaluation sweeps, and multi-seed verification—demonstrates that high-quality scientific Q&A systems can be developed at modest cost using a single cloud A100 instance. The cfg5 Full Agent evaluation (\$20.97 alone) is disproportionately expensive because self-consistency inference ($N = 5$) multiplies the evaluation time roughly 4×, yielding 342.8 minutes for 217 samples.

F. SUMMARY OF KEY FINDINGS

1) Curriculum SFT is the primary driver of improvement. cfg1 (SFT Phase 1) adds +5.53 pp over zero-shot; adding the logic curriculum phase (cfg2) adds a further +2.77 pp. Together, SFT accounts for +8.30 pp of total gain.



Error distribution for cfg3 ($n_{\text{wrong}} = 103$). Logical fallacy errors (E4) dominate, driven entirely by the logic sub-domain where formal quantifier reasoning remains weak. Unit/dimension errors (E1) constitute the second-largest category within the physics sub-domain.

physics problems phrased as logical conditionals) that load the wrong adapter.

- Self-consistency provides no benefit at 4B scale. cfg5 (50.69%) equals cfg1 despite adding five-sample SC and domain routing. Errors at 4B scale are primarily systematic (the model consistently misidentifies a quantifier) rather than random, so majority voting reinforces the systematic error.
- Logic reasoning remains a significant bottleneck. Peak logic accuracy is 30.26% (cfg1 and cfg4) versus 68.09% physics. All 57 logic errors in cfg3 are classified as E4 (logical fallacy), suggesting this requires either larger model capacity, process-reward supervision, or formal deduction modules.
- Unit/dimension errors motivate the unit reward. E1 (unit/dimension errors) constitute 25.2% of total cfg3 failures and 100% occur on physics problems. The zero E2 (wrong formula) count confirms the model selects correct physical laws but sometimes mis-scales the output by a dimensional factor.

Compute cost breakdown (one NVIDIA A100-SXM4-80GB, \$3.67/GPU-hr).

Phase / Activity	Time (min)	USD
Training		
SFT Phase 1	58.8	\$3.60
SFT Phase 1.5 (logic)	15.7	\$0.96
GRPO Phase 2 (mixed)	79.8	\$4.89
GRPO Phase 2P (phys)	76.6	\$4.69
GRPO Phase 2L (logic)	62.9	\$3.85
Subtotal training	293.8	\$17.99
Evaluation (6 configurations, 217 samples each)		
cfg0 (zero-shot)	63.9	\$3.91
cfg1 (SFT Ph.1)	85.6	\$5.24
cfg2 (Logic SFT)	84.0	\$5.14
cfg3 (GRPO-mixed)	84.3	\$5.16
cfg4 (Dual-LoRA)	90.2	\$5.51
cfg5 (Full Agent)	342.8	\$20.97
Subtotal eval	750.8	\$45.93
Grand total	1,044.6	\$63.89

the same dimensional factor across all $N = 5$ sampled completions. Majority voting over five identical wrong answers yields the same wrong answer with higher confidence, explaining the absence of SC gains. This finding is consistent with the observation that SC improves primarily at larger model scales (e.g., $> 30B$ parameters in the original paper [2]) where the diversity of reasoning paths is sufficient for the ensemble effect to act.

B. WHY DOES GRPO PROVIDE ONLY MARGINAL IMPROVEMENT?

The marginal GRPO gains (cfg3 $\approx$ cfg2) suggest that the Phase 1.5 SFT checkpoint has already captured the primary task-relevant patterns, and 250 GRPO steps at the low learning rate 10^{-6} with $K = 4$ completions may be insufficient to drive substantial additional policy improvement. Additionally, GRPO’s reward signal relies on correctness of the extracted answer; when the model is systematically wrong (E4: logical fallacy, 55.3% of errors), all $K = 4$ completions receive near-zero reward within each group, making the normalised relative reward $\hat{r}_k$ uninformative.

C. THE UNIT-AWARE REWARD: DESIGN RATIONALE AND LIMITATIONS

The unit reward r_{unit} is designed to encourage dimensional consistency in physics answers. Our error analysis shows that 25.2% of failures are unit/dimension errors (E1), supporting the hypothesis that this signal is meaningful. However, the reward ablation experiment (Step 5) could not be completed due to a TRL 1.5.1 API incompatibility (the completions argument changed from plain strings to structured message dictionaries). Consequently, we cannot report isolated causal evidence for the unit reward’s contribution to the observed physics accuracy gains. This remains an open experimental question for future work.

D. DUAL-LORA ROUTING: PROMISE AND FAILURE MODES

The Dual-LoRA architecture is theoretically well-motivated: physics and logic problems require different inductive biases, and dedicated LoRA adapters trained on domain-specific objectives should specialise accordingly. In practice, the TF-IDF + logistic regression router achieves $\sim 90\%$ classification accuracy on clean domain labels, but the 10% of misclassified questions are disproportionately hard—often physics problems phrased as logical conditionals (e.g., “if the voltage increases, what happens to current?”) or logic problems involving quantitative premises. These border-zone questions are precisely the most difficult ones, and routing errors on them load the wrong adapter, compounding the error. A neural routing layer trained jointly with the adapters, or an ensemble approach, may mitigate this issue.

E. LIMITATIONS

- Evaluation set size: 217 samples yield 95% confidence intervals of approximately ± 3.3 pp, limiting statistical power for differences < 2 pp.
- Single competition domain: generalisation to established science benchmarks such as SciBench [10] and GPQA [20] is not assessed.
- Incomplete reward ablation: the Step 5 reward ablation failed due to a TRL API incompatibility, leaving the causal contribution of r_{unit} unconfirmed.
- No process-level reward: step-annotated data could enable Process Reward Models [4] for finer-grained guidance, potentially addressing the systematic logical errors.
- Short GRPO training: 250 steps may be insufficient for convergence; longer GRPO training on a larger compute budget is a natural future direction.

We presented TRA-SAE, a training pipeline for mixed-domain scientific reasoning that combines curriculum supervised fine-tuning (SFT Phase 1 on all 1,945 samples, Phase 1.5 logic specialisation) with unit-aware Group Relative Policy Optimization (GRPO) and an optional Dual-LoRA expert routing agent.

Our best configuration (cfg2: curriculum SFT) achieves **53.46%** overall accuracy on the EXACT 2026 validation set—an +8.30 pp improvement over our zero-shot baseline and +14.75 pp above the competition public baseline—while the GRPO variant (cfg3) achieves a comparable $53.15\% \pm 1.15\%$ across three seeds. Both improvements are statistically significant ($p < 0.05$, McNemar’s test). The entire experimental pipeline costs \$63.89 in GPU compute on a single A100-80GB.

Key contributions are: (i) the unit-aware reward, which targets the 25.2% of physics failures attributable to dimensional inconsistency; (ii) evidence that curricu-

lum logic SFT (+2.77 pp) is more efficient than GRPO for the same compute budget; (iii) informative negative results showing that self-consistency and Dual-LoRA routing degrade performance at 4B scale.

Future work will investigate (i) scaling to 7B/14B checkpoints for both the GRPO and Dual-LoRA configurations; (ii) process-level reward models trained on step-annotated logic derivations to address the dominant E4 logical-fallacy error category; (iii) completion of the reward ablation experiment under $\text{TRL} \geq 1.5.2$ to quantify the isolated contribution of r_{unit} ; and (iv) cross-dataset evaluation on SciBench [10] and GPQA [20] to assess out-of-domain generalisation.

- [1] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24824–24837, 2022. [Online]. Available: <https://arxiv.org/abs/2201.11903>
- [2] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," *arXiv preprint arXiv:2203.11171*, 2023. [Online]. Available: <https://arxiv.org/abs/2203.11171>
- [3] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in Neural Information Processing Systems*, vol. 35, pp. 22199–22213, 2022. [Online]. Available: <https://arxiv.org/abs/2205.11916>
- [4] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, "Let's verify step by step," *arXiv preprint arXiv:2305.20050*, 2023. [Online]. Available: <https://arxiv.org/abs/2305.20050>
- [5] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo, "DeepSeekMath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.03300>
- [6] A. Yang, B. Zhang, B. Hui, B. Gao, B. Yu, C. Li, D. Liu, J. Tu, J. Zhou, J. Lin, K. Lu, M. Xue, R. Lin, T. Liu, X. Ren, and Z. Zhang, "Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement," *arXiv preprint arXiv:2409.12122*, 2024. [Online]. Available: <https://arxiv.org/abs/2409.12122>
- [7] Z. Azerbayev, H. Schoelkopf, K. Paster, M. D. Santos, S. McAleer, A. Q. Jiang, J. Deng, S. Biderman, and S. Welleck, "Llemma: An open language model for mathematics," *arXiv preprint arXiv:2310.10631*, 2024. [Online]. Available: <https://arxiv.org/abs/2310.10631>
- [8] L. Yu, W. Jiang, H. Shi, J. Yu, Z. Liu, Y. Zhang, J. T. Kwok, Z. Li, A. Weller, and W. Liu, "MetaMath: Bootstrap your own mathematical questions for large language models," *arXiv preprint arXiv:2309.12284*, 2023. [Online]. Available: <https://arxiv.org/abs/2309.12284>
- [9] X. Yue, X. Qu, G. Zhang, Y. Fu, W. Huang, H. Sun, Y. Su, and W. Chen, "MAmmoTH: Building math generalist models through hybrid instruction tuning," *arXiv preprint arXiv:2309.05653*, 2023. [Online]. Available: <https://arxiv.org/abs/2309.05653>
- [10] X. Wang, Z. Hu, P. Lu, Y. Zhu, J. Zhang, S. Subramaniam, A. R. Loomba, S. Zhang, Y. Sun, and W. Wang, "SciBench: Evaluating college-level scientific problem-solving abilities of large language models," *arXiv preprint arXiv:2307.10635*, 2024. [Online]. Available: <https://arxiv.org/abs/2307.10635>
- [11] L. Pan, A. Albalak, X. Wang, and W. Y. Wang, "Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning," *arXiv preprint arXiv:2305.12295*, 2023. [Online]. Available: <https://arxiv.org/abs/2305.12295>
- [12] T. X. Olausson, A. Gu, B. Lipkin, C. E. Zhang, A. Solar-Lezama, J. B. Tenenbaum, and R. P. Levy, "LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers," *arXiv preprint arXiv:2310.15164*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.15164>
- [13] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, and G. Neubig, "PAL: Program-aided language models," *arXiv preprint arXiv:2211.10435*, 2022. [Online]. Available: <https://arxiv.org/abs/2211.10435>
- [14] P. Wang, L. Li, Z. Shao, R. X. Xu, D. Dai, Y. Li, D. Chen, Y. Wu, and Z. Sui, "Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations," *arXiv preprint arXiv:2312.08935*, 2024. [Online]. Available: <https://arxiv.org/abs/2312.08935>
- [15] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2022. [Online]. Available: <https://arxiv.org/abs/2106.09685>
- [16] C. Huang, Q. Liu, B. Y. Lin, T. Pang, C. Du, and M. Lin, "LoRAHub: Efficient cross-task generalization via dynamic LoRA composition," *arXiv preprint arXiv:2307.13269*, 2023. [Online]. Available: <https://arxiv.org/abs/2307.13269>
- [17] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. Christiano, J. Leike, and R. Lowe, "Training language models to follow instructions with human feedback," *Advances in Neural Information Processing Systems*, vol. 35, pp. 27730–27744, 2022. [Online]. Available: <https://arxiv.org/abs/2203.02155>
- [18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," in *arXiv preprint arXiv:1707.06347*, 2017. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [19] R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: <https://arxiv.org/abs/2305.18290>
- [20] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman, "GPQA: A graduate-level google-proof q&a benchmark," *arXiv preprint arXiv:2311.12022*, 2023. [Online]. Available: <https://arxiv.org/abs/2311.12022>

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